COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	8/16/26	**DATE SUBMITTED:**	8/20/26
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ACTIVITY NO. 6.1 Linear and Binary Search

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the following questions:

1. What is a search tree in data structures?

A process of locating a specific node or key within a tree data structures

2. What are the different types of search algorithms in data structures? Differentiate each type of search.

1. Linear Search
Iterates through data elements one by one from start to end until a match is found or the dataset ends

2. Binary Search
Compares target to the middle element of a sorted list and halves the remaining search space each step

3. Ternary Search
Divides a sorted search space into three equal parts using two midpoints to narrow down the target location

4. Jump Search
Skips ahead through a sorted array by fixed block steps of size square root of n, then performs a linear search within the identified block

5. Interpolation Search
Estimates the target index in a uniformly distributed sorted array based on the values at the current boundaries

6. Exponential Search
Finds a target range by exponentially doubling the search index, then executes a binary search within that bounded range

7. Breadth First Search

Traverses a tree or graph level by level across nodes using a first in first out queue structure

8. Depth First Search
Traverses a tree or graph as far down a single branch as possible before backtracking using a fast In first out stack or recursion

9. Binary Search Tree
Traverses node branches by comparing key values, moving to the left child for smaller values and to the right child for larger values

10. Hash Lookup
Passes a search key into a mathematical hash function to generate a direct memory index for constant amount of time retrieval

11. Knuth Morris Pratt
Searches for sub patterns in text by utilizing a precomputed longest prefix suffix table to skip redundant comparisons

12. Rabin Karp
Calculates and compares numeric hash values of sub patterns within a continuous text sequence before doing direct character checks

3. What operations / implementations can be performed using binary and linear search operations?

Binary and linar is used to locate elements, validate conditions, and solve computational problems efficiently

4. What are the advantages in using binary search trees as data structure?

Faster searching and faster input and output

5. Give an example program using binary search and Linear search.

Contacts in phones for linear and microsoft word searcher for binary

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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